After completing the preparation materials, students should be able to:

- Give examples of how each of the pharmacokinetic properties of the antiviral agents contributes to patient care.
- Describe the mechanisms of action of each class in the context of the functions of the targeted viral biomolecules.
- Relate viral resistance to the antiviral mechanisms of action.
- Explain the goals of therapy and evidence-based rationales for the clinical selection of specific antiviral drugs.
- List the potential adverse effects and drug interactions.

Key points: Influenza Antiviral Agents

- Influenza antiviral agents are important adjuncts to influenza vaccine in the control of influenza.
- They are used for treatment of acute uncomplicated influenza, post-exposure prophylaxis in patients at risk of influenza complications, and pre-exposure prophylaxis in widespread influenza outbreaks for unvaccinated individuals at very high risk of developing complications, such as immunocompromised patients.
- When initiated within 48 hours of the onset of symptoms, treatment can lessen symptoms, shorten illness duration by 1 or 2 days, decrease duration of viral shedding, and may reduce the risk of lower respiratory tract complications and hospitalizations.
- The different agents have various routes of administration, half-lives, and indications for use.
- Prophylaxis does not eliminate risk of influenza; susceptibility returns when the drug is removed. Adjunctive vaccine should be administered.

Key points

- The neuraminidase inhibitors (NAIs), oseltamivir, zanamivir, and peramivir, prevent viral cleavage of sialic acid residues from viral proteins, which prevents the release of new influenza A and B virions. Oseltamivir and zanamivir are neuraminic acid analogs. Peramivir was designed to have high affinity binding to the neuraminidase active site. Viral resistance to the NAIs is rare.
- **Oseltamivir** is the preferred agent for treatment and prophylaxis of influenza in outpatients and hospitalized patients. It is typically taken orally twice daily x5 days for treatment. It can cause nausea and vomiting. Taking with food may improve tolerance.
- **Zanamivir** for oral inhalation twice daily x5 days (treatment) is an alternative to oseltamivir. It can cause bronchospasm and should be avoided in patients with underlying respiratory disease. Allergic reactions are reported.
- **Peramivir** is an alternative to oseltamivir for treatment only. It is not indicated for prophylaxis. It has a long half-life and is given intravenously x1 dose for outpatients and hospitalized patients. Adverse effects include hyperglycemia, skin reactions, hypersensitivity reactions.
- NAIs are associated with transient neuropsychiatric events – confusion, delirium, hallucinations.
- Oseltamivir is the preferred drug for pregnant individuals. Zanamivir and peramivir are alternatives. All NAIs may be administered to pediatric patients (drug-specific age recommendations).

Key points

- **Baloxavir** is an influenza cap-dependent endonuclease (PA protein) inhibitor effective against influenza A and B. Point mutations in viral PA protein confer resistance. It is administered orally x1 dose for treatment and post-exposure prophylaxis. Administration along with polyvalent cations (foods, oral supplements, and bile acid sequestrants) should be avoided. Baloxavir is generally well tolerated. Baloxavir is not currently recommended in pregnancy due to lack of safety and efficacy data.
- **Adamantanes, amatadine and rimantadine**, prevent viral uncoating and replication by blocking the M2 protein ion channel of influenza A, but not influenza B. Adamantanes are not recommended for influenza treatment or prophylaxis because of high-level resistance by A/H1N1 and A/H3N2 viruses.
- Amantadine distributes to the CNS and can cause cognitive effects, agitation, hallucinations, and others. It is used as an adjunct in Parkinson disease treatment. Rimantadine has a low incidence of CNS effects. They are both taken orally.

Therapeutic uses of the anti-influenza class of drugs

Treatment of acute uncomplicated influenza
Start within 48 hours of the onset of signs / symptoms
Treatment can:
- Lessen symptoms
- Shorten illness duration by 1 or 2 days
- Decrease duration of viral shedding
- May reduce the risk of lower respiratory tract complications and hospitalizations in adults and children compared to placebo or no treatment
- oseltamivir, zanamivir, peramivir, baloxavir

Prophylaxis: Vaccination

Post-exposure Prophylaxis: Administer within 48 hours of exposure
- Individuals at high risk for severe or complicated influenza
- Healthy unvaccinated people who have close contact with individuals at high risk of complication of influenza
- oseltamivir, zanamivir, or baloxavir

Pre-exposure Prophylaxis:
During widespread outbreaks, consider for unvax. pts at very high risk of developing complications
oseltamivir, zanamivir

Neuraminidase Inhibitors Pharmacokinetic Properties

Takeaways: “Favorable” PK profile		Oseltamivir	Zanamivir	Peramivir
Adherence to therapy	A	Oral prodrug Tx: 2x/d x5 days typically PPx: 1x/d x7-14d after exposure	Oral inhalation (powder) Low systemic absorption Tx: 2x/d x5 days PPx: as for oseltamivir	IV x 1 dose Hospitalized patients: Tx:1x/d x5-10 days (*oseltamivir preferred*)
	D	Low to moderate plasma protein binding free drug to tissues to site of action		
No metabolism-related drug interactions	M	Esterase hydrolysis to active carboxylate Bioavailability ~75%	Minimal metabolism	Minimal metabolism
	E	Renal, OAT1/3 secretion of parent and active metabolite	Urine, unchanged (less than 8% of dose)	Renal, unchanged drug
Renal dosing		Dose adjustment for renal impairment		Dose adjustment for renal impairment
Dose frequency	t½	6-10 h (active metabolite)	Up to 5 h (serum)	20 h
		All NAIs may be used in pediatrics, adolescents, adults, and pregnancy.		

Therapeutic uses of NAIs

drug of choice	alternatives to oseltamivir	
Oseltamivir	Zanamivir	Peramivir
Treatment and prophylaxis of infants, children, adoles cents, and adults outpatients and hospitalized patients	Treatment, outpatients: children ≥7 years old and adults Prophylaxis in children ≥5 years and adults • Not recommended in patients with underlying respiratory disease (eg asthma, COPD)	Treatment, outpatients: children ≥2 years old and adults Treatment, hospitalized pts: Alternative to oseltamivir if the patient cannot tolerate or absorb oseltamivir Not used for prophylaxis

- Oseltamivir is the preferred drug for the treatment and prophylaxis of influenza in pregnancy.

Zanamivir and peramavir are alternatives if oseltamivir is not tolerated.

- Neuraminidase Inhibitors Potential Adverse Effects
- These agents are generally well tolerated.

Oseltamivir	Zanamivir	Peramivir
Most common: • Nausea and vomiting May be taken with food to improve tolerance • Headache • Serious skin reactions	• Headache • Bronchospasm risk in patients with underlying airways disease • Sinusitis; dizziness; ear, nose, throat infections • Allergic reactions: oropharyngeal or facial edema Not recommended for patients with underlying respiratory disease (eg asthma, COPD)	• Diarrhea or constipation • Hyperglycemia • Serious skin reactions • Hypersensitivity reactions
NAIs Neuropsychiatric events, rare, transient (post-marketing surveillance) Confusion, delirium, hallucinations potentially with behavioral or neurologic changes Occur primarily in children and adolescents. The cause has not been established.		

Baloxavir marboxil Pharmacokinetics

only one dose needed	A	Oral prodrug **Single dose** within 48 hours of symptom onset reduces symptoms by approximately 1-2 days
low potential for metabolism, transport DDIs	D	Very large Vd (1,180 L) minor P-glycoprotein substrate Vd helped inform the manufacturer's dosing recommendations. • As weight increases, exposure to baloxavir decreases higher dose for patients with a higher BMI
	M	UGT1A3 conversion to active metabolite, baloxavir (minor CYP3A4)
renal dose adjust. likely not needed only one dose needed	E t½	Feces (80%) Urine (15%; 3% as baloxavir) ~80 hours

Baloxavir Effects

- Activity: Effective against influenza A and B
- Resistance may emerge – point mutations in PA protein
- Therapeutic use:
  - Acute uncomplicated influenza and post-exposure prophylaxis in patients ≥12 years at high risk of developing complications
  - Children 5 years who are otherwise healthy
  - NOT recommended in pregnancy – lack of safety and efficacy data
- Adverse effects: Generally well tolerated
  Low incidence of side effects:
  - headache
  - diarrhea
  - nasopharyngitis
- Drug interactions: polyvalent cations
  - Chelate baloxavir absorption of, thus, exposure of the body to the drug
  - Avoid calcium-containing foods and cation-containing supplements

Adamantanes Pharmacokinetic Properties

	Amantadine	Rimantadine
A	Oral Bioavailability ~90%	Oral Well absorbed
D	Distributes to CNS ***Elderly are more sensitive to amantadine's CNS adverse effects.***	Low CNS distribution Moderate protein binding
M	Minimal	Extensive hepatic hydroxylation, conjugation, glucuronidation
E	Urine, ~90% unchanged	Urine, metabolites
	Dosage adjustment recommended in severe renal or hepatic impairment.	
t½	12-18 hours: young adults 29 hours: healthy older adults 8 days: end stage renal disease (not dialyzable)	13-65 hours: adults 20-65 hours: elderly

Prophylaxis and Treatment of Acute Uncomplicated Influenza

Neuraminidase Inhibitors

Competitive inhibition of the enzyme viral clumping interferes with release of virus progeny

Oseltamivir preferred agent	Zanamivir	Peramivir
Oral	Oral inhalation	I.V.
All ages	Pts 7 yrs	Pts 6 mon
BID x5d	BID x5d	1 dose
Nausea	Bronchospasm	Skin reactions

Neuropsychiatric adverse effects (rare)
Confusion, delirium, hallucinations
May be associated with behavioral or neurologic changes

Cap-dependent Endonuclease Inhibitor

Baloxavir inhibits host mRNA “cap snatching” inhibits initiation of RNA synthesis

Baloxavir marboxil

Oral

Pts 5 yrs

1 dose

Diarrhea, Hypersensitivity

Chelates polyvalent cations

Amantadine, Rimantadine: M2 proton ion channel inhibitors prevent uncoating/assembly, influenza A only
Not recommended – Resistance H1N1, H3N2

Key points: COVID-19 Antiviral Therapy

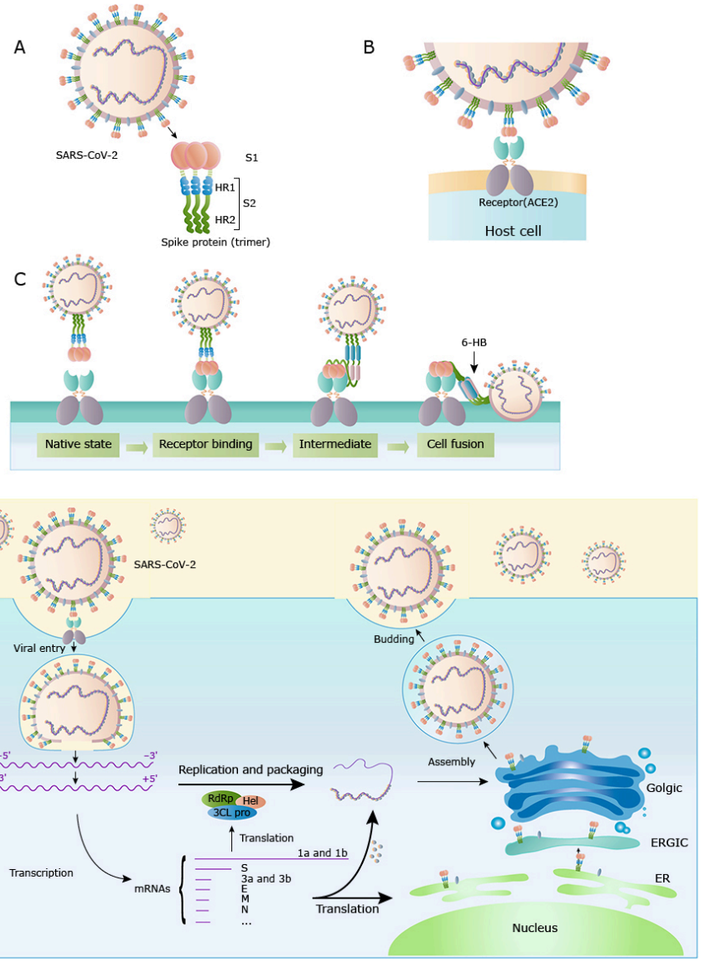
- To determine the appropriateness of COVID-19-specific antiviral therapy, patients are evaluated for suspected or confirmed COVID-19 to assess the severity of symptoms, risk of progression to severe disease, and safety of the home setting, with shared clinical decision making with the patient.
- Outpatient treatment of acute illness is recommended for symptomatic patients with mild to moderate illness and risk factors for progression to severe disease, such as older age, immune status, COVID-19 vaccination history, or comorbidities associated with progression.
- **Nirmatrelvir-ritonavir** is the recommended agent. Nirmatrelvir is a peptidomimetic inhibitor of SARS-CoV-2 3CLPRO (same as main protease MPRO), which prevents processing of viral polyprotein precursors, thus, inhibiting viral replication. Ritonavir inhibits CYP3A4, which increases nirmatrelvir concentrations. The drug combination is initiated as soon as possible within 5 days of symptom onset and administered twice daily for five days. Studies have shown that it substantially reduces the risk of hospitalization and mortality in high-risk individuals with mild to moderate COVID-19. It is FDA approved for adults, adolescents, and children 12 years old. Renal dosing is listed in the drug monograph. There is a high potential for drug interactions.
- **Ensitrelvir** (FDA approved May 2026) is a SARS-CoV-2 3CLPRO inhibitor. It has a long half-life and does not need a PK booster.

Key points

- **Remdesivir** is an adenosine nucleotide analog activated by triphosphorylation, which competes for incorporation into viral RNA chains by RNA-dependent RNA polymerase (RdRp), resulting in delayed chain termination. Remdesivir is FDA approved for the treatment of mild to moderate COVID-19 in non-hospitalized and hospitalized infants 3 kg, children, adolescents, and adults. The doses are once-daily intravenous infusions x3 days in non-hospitalized patients and 5 days usually or until discharge for hospitalized patients (not on mechanical ventilation). The drug reduces COVID-19 hospitalizations in acute symptomatic patients with mild to moderate illness and may reduce time to recovery and risk of mechanical ventilation in hospitalized patients with severe COVID-19 not on mechanical ventilation.
- **Molnupiravir** has US Emergency Use Authorization (EUA) for outpatient treatment of mild to moderate symptomatic patients 18 years old who are at high risk of progression to severe disease. Potential benefit is likely modest. Molnupiravir did not reduce the risk of hospitalization but did reduce the time to recovery. The drug is metabolized to cytidine nucleoside analog, NHC. Triphosphorylated NHC is incorporated into SARS-CoV-2 RNA viral RdRp. When RdRp copies the RNA strand containing NHC, mutations accumulate, resulting in errors in the viral genome – lethal mutagenesis (viral error catastrophe) – inhibiting replication. The drug is initiated as soon as possible within 5 days of symptom onset and administered twice daily for five days.
- **Pemivibart** is an IgG1 antibody directed against SARS-CoV-2 spike protein for prophylaxis of COVID-19 in patients ≥12 years old weighing ≥40 kg with moderate to severe immune compromise who cannot be vaccinated or have inadequate response to vaccination (US Emergency Use Authorization).

SARS-CoV-2 Replication

Structure and function of the SARS-CoV-2 spike protein



(A) The schematic structure of the spike (S) protein.

(B) The spike (S) protein binds to the receptor ACE2.

(C) The binding and virus-cell fusion process mediated by the spike (S) protein.

(D) The life cycle of SARS-CoV-2 in host cells.

SARS-CoV-2: severe acute respiratory syndrome coronavirus 2.

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- The primary host cell entry receptor for SARS-CoV-2 is angiotensin-converting enzyme 2 (ACE2). SARS-CoV-2 binds ACE2 through the receptor-binding domain of its spike protein and together with host factors the virus fuses with the cell's outer membrane and enters the cell by endocytosis.
- After uncoating, the viral positive-sense RNA uses the cell's machinery to produce structural proteins necessary for assembling new virus particles and to produce complementary negative-sense RNA to synthesize more positive-stranded RNA genomes. The new viruses are transported by the Golgi apparatus to the cell membrane where they are released from the host cell by exocytosis or upon cell lysis.

Variants:

Additional enzymatic interactions between SARS-CoV-2 and host cells determine specific cell tropism, which has evolved with the virus.

Following ACE2 binding, the original SARS-CoV-2 virus and some subsequent variants, including Delta, enter cells after a host protease (most often the TMPRSS2 enzyme which is expressed in high levels in lung epithelial cells) cleaves the spike protein, enabling subsequent membrane fusion and cell entry.

Omicron-lineage variants, in contrast, mostly enter host cells through a different pathway involving cathepsin-dependent endocytosis and use the TMPRSS2-mediated pathway far less efficiently. Thus, Omicron-lineage variants replicate more efficiently in the upper airways, where TMPRSS2 expression is lower than in the lungs.

Mechanisms of Action: All these agents are effective against the ancestral virus and the other variants that cause illness.

Nirmatrelvir Protease inhibitor	**Remdesivir** RdRp inhibitor	**Molnupiravir** “Viral error catastrophe”
Inhibits 3CLPRO, 3CLPRO cleaves SARS-CoV-2 polyproteins, 1a and 1b Inhibits the release of nonstructural proteins, including proteases Inhibits viral replication	Remdesivir adenosine nucleotide intracellular phosphorylation remdesivir-triphosphosphate (TP) competes with adenosine-TP for incorporation into RNA chains by RdRp chain termination after incorporation of a few more nucleotides delayed chain termination during viral replication	**Step 1.** NHC intracellular phosphorylation active NHC-TP **Step 2.** RdRp takes up NHC-TP as a substrate instead of cytosine-TP and uridine-TP. **Step 3.** When RdRp copies the RNA strand containing NHC, more mutations are incorporated, which leads to mispairing and *accumulation of mutated RNA* eventually lethal to the virus. **Result:** lethal mutagenesis “viral error catastrophe”
Ritonavir: Pharmacokinetic enhancer CYP3A4 inhibitor		
Ensitrelvir 3CLPRO inhibitor Does not require PK booster		

RdRp: RNA-dependent RNA polymerase

Therapeutic Use: Administer as soon as possible after onset of symptoms and within 5 days of symptom onset.

	Nirmatrelvir with ritonavir	**Remdesivir**	**Molnupiravir**
Use	Mild to moderate illness Outpatients at high risk for progression to severe COVID-19 or hospitalization ***Preferred agent***	Mild to moderate illness Outpatients at high risk for progression to severe COVID-19* Hospitalized patients not on mechanical ventilation.	Mild to moderate illness Outpatients at risk for severe COVID-19 or hospitalization Alternative agent
	Adults Adolescents 12 years and 40 kg	Adults Infants 28 days, 3 kg Children and Adolescents	Adults Adolescents 18 years and 40 kg
Pregnancy	Treatment recommended if indicated.	Treatment recommended if indicated.	Not recommended. Patients who may become pregnant should use reliable contraception during therapy and for 4 days after the last dose.
	*Remdesivir may be administered only in health care settings where clinicians have immediate access to medications for treating severe infusion reactions or hypersensitivity reactions.		

Adverse Effects (not a complete list)

Remdesivir	• Severe bradycardia (mechanism not known) • Increased serum aminotransferases, reversible; acute hepatic failure • Hypersensitivity reactions, anaphylaxis, angioedema • Hyperglycemia • Injection site erythema Administer only in health care settings where clinicians have access to medications for severe infusion reactions or hypersensitivity reactions. Drug interactions: • Chloroquine/hydroxychloroquine may decrease therapeutic effectiveness of remdesivir – avoid combination
Molnupiravir	• Skin rash • Hypersensitivity reactions, anaphylaxis, angioedema Not recommended for pregnant or pediatric patients. • Molnupiravir may cause harm to the fetus (animal studies: teratogenicity / fetal loss) • Molnupiravir is not recommended in patients <18 years old due to potential for affects on bone and cartilage growth (seen in animal studies)

Questions: You may want to pause the video to think about the answers.

- A 10-year-old child is brought to the pediatrician with abrupt onset of fever, nonproductive cough, and complaints of muscle pains 3 hours ago. Rapid antigen test is positive for the circulating virus. The patient is prescribed an antiviral oral suspension that prevents budding of the viral progeny from host cells. The mother is directed to give the child 10 mL of the medicine by mouth every 12 hours for 5 days (10 doses) with food to reduce stomach upset. What is the drug?
- Oseltamivir – metabolized to active oseltamivir carboxylate. Influenza characteristically begins with an abrupt onset of fever, nonproductive cough and myalgia.
- A 68-year-old male with a history of hypertension and diabetes presents to the family practice clinic with complaint scratchy throat, headache, and muscle aches that started yesterday and have progressively worsened. Home COVID testing was positive. After checking for significant drug-drug interactions, the doctor submits an e-prescription for the preferred oral antiviral agent. What is it?
- Nirmatrelvir with ritonavir, the preferred agent.

Questions: Do you want to pause the video?

- A 14-year-old person (pronouns she/her/hers) with a history of severe asthma has a telemedicine visit with her pulmonologist to report that she was exposed to influenza at school yesterday and at her friend's house today. She reports that she is feeling well and has no symptoms. The patient received her flu vaccine 7 days ago. She had an allergic reaction to the oral NAI last year. An e-prescription is submitted for one dose of an oral antiviral agent having a unique mechanism of action and a long half-life. She is advised not take to it at the same time as dairy products or vitamin and mineral supplement. What is the mechanism of action of the antiviral agent?
- Baloxavir marboxil is converted to active baloxavir, a cap-dependent endonuclease inhibitor that blocks influenza A and B proliferation by inhibiting the host cellular pre-mRNAs that serve as primers for viral mRNA. This action prevents initiation of viral mRNA synthesis.
- A 77-year-old male with dyspnea and a positive COVID-19 test is hospitalized for treatment with an intravenous antiviral agent. What is the mechanism of the drug?
- The drug is remdesivir, the only one of the currently available SARS-CoV-2 antiviral agents for the treatment of hospitalized patients not on a mechanical ventilator. It is an adenosine nucleotide analog triphosphorylated and taken up by RdRp. It causes delayed termination of RNA synthesis.

Small molecule drugs that inhibit SARS-CoV-2 replication by different mechanisms

Nirmatrelvir-ritonavir preferred agent	Remdesivir	Molnupiravir
3CLPRO main viral protease inhibitor	RdRp inhibitor Delayed chain termination	Lethal mutagenesis (viral error catastrophe) by uptake NHC-TP
Oral, CYP3A4 Ritonavir booster	IV infusion	Oral
Pts 12yrs; 40kg	Pts 28 days old outpatient or hospitalized	Pts 18yrs; 40kg
BID x5d	Daily x3d or 5d	BID x5d
Nausea, diarrhea Dysgeusia Ritonavir-related	Bradycardia Skin reactions	Potential for fetal harm, bone/cartilage growth in peds <18yo (animal studies)

Nirmatrelvir with ritonavir: Mild to moderate illness; outpatients at risk for severe COVID-19 or hospitalization
Remdesivir: Outpatients with mild to moderate illness at high risk for progression to severe COVID-19 (IV daily x3 days) and hospitalized patients not on mechanical ventilation (IV daily x5 days usually)
Molnupiravir: Mild to moderate illness; outpatients at risk for severe COVID-19 or hospitalization
Pemivibart: Single IV dose for p rophylaxis of COVID-19 for patients ≥12 years old weighing ≥40 kg with moderate to severe immune compromise who cannot be vaccinated or have inadequate response to vaccination
Administration of remedesivir and pemivibart only in health care settings where clinicians have access to medications for severe infusion reactions and hypersensitivity reactions.

Questions and Answers

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